

Tethered swimming, a tool for measuring propulsive force in an aquatic context

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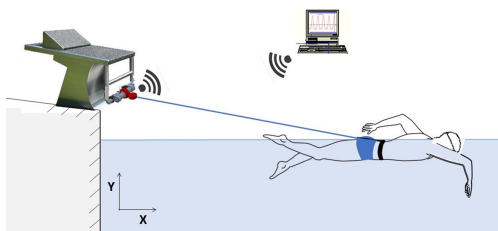
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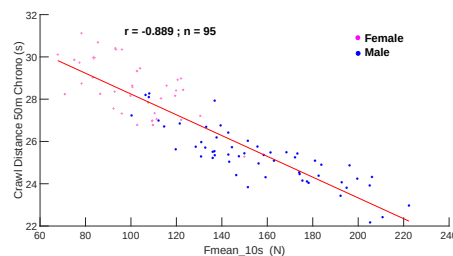
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From a biomechanical point of view, propulsive force is a major component of swimming performance. Swimmers' forces evaluation is therefore an important issue for both coaches and scientists. Among the methods for measuring force in the water, tethered swimming has been recognized as a simple, relevant and reliable experimental method [1]. Although tethered swimming is not exactly similar to free swimming [2] it can be used to measure a swimmer's propulsive force in an aquatic context. As part of the D-Day project (PPR: Priority Research Program "Very high performance sport"), we used this method to assess the propulsive force of swimmers and determine the most relevant dynamic parameters for predicting swimmer's performance.



Tethered swimming force measurement device



Mean force vs 50m freestyle performance

Eleven dynamic parameters taken from the literature, and two test durations, were studied to determine which test was most predictive of performance. 178 expert swimmers completed these tethered tests (71 women and 107 men, 18.1 ± 3.8 years old, height: 1.79 ± 0.09 m, mass: 68.1 ± 11.3 kg, level: 566.6 ± 87.3 pts of FINA Point Scoring 2022). The results show that the mean force in 10s test is the parameter that correlates most with 50m ($r=-0.889$) and 100m ($r=-0.861$) freestyle performances. Maximum force in 30s test was the parameter that correlates most with 200m ($r=-0.773$) freestyle performances. Over 400m freestyle, all the parameters showed a moderate correlation ($r \leq -0.616$) with the performance. So, the mean force over a 10s test duration should be the parameter and conditions most commonly used by coaches to assess swimmers' propulsive force in the water over 50m and 100m events. Maximum force and duration of 30s test should be used to evaluate propulsive force for 200m events. Beyond 200m, it seems necessary to increase the duration of the tests in tethered swimming to assess the propulsive force.

References

- [1] Amaro N., Morouço P., Marques M., Fernandes R. & Marinho D. J hum sport exerc **12**(4), 1346-1360 (2017).
- [2] Samson M., Monnet T., Bernard A., Lacouture P. & David L. Sport Biomech **18**, 571–586 (2019).